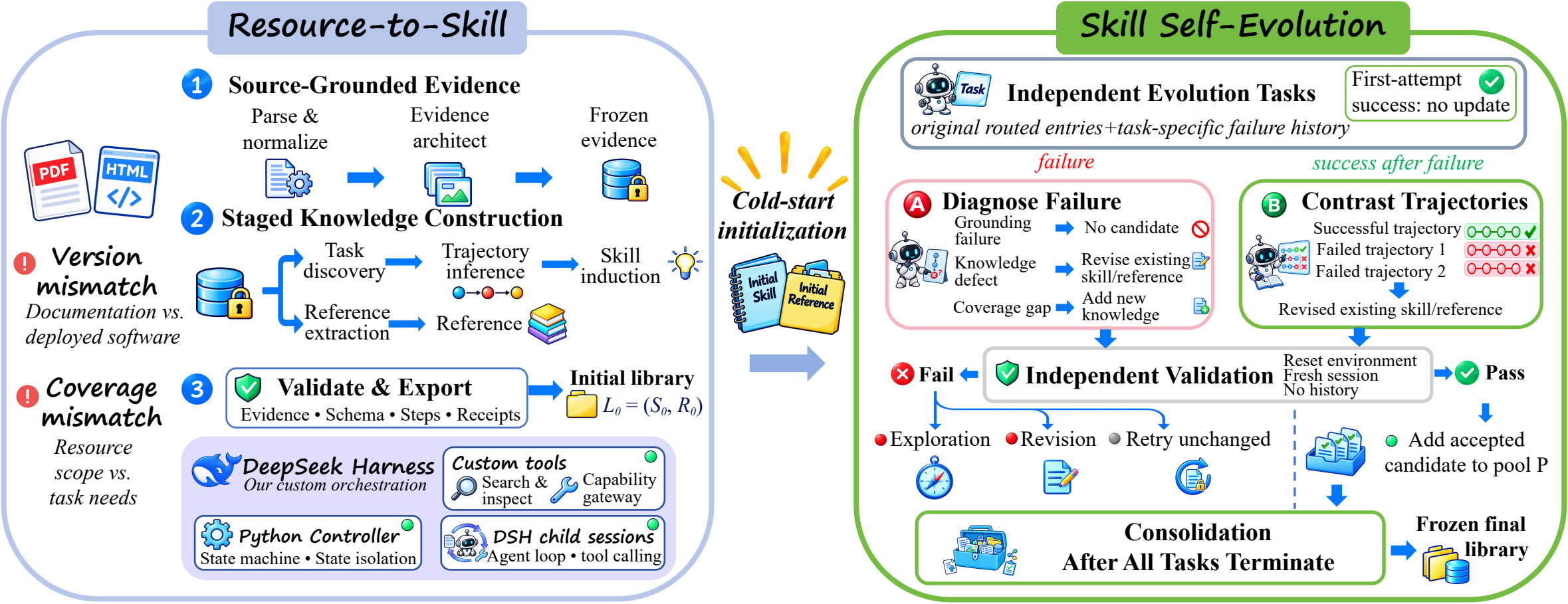

Resource-to-Skill
1 Source-Grounded Evidence
Parse & normalize
Evidence architect
Frozen evidence
PDF
HTML
2 Staged Knowledge Construction
Task discovery
Trajectory inference
Skill induction
Reference extraction
Reference
Version mismatch
Documentation vs. deployed software
Coverage mismatch
Resource scope vs. task needs
3 Validate & Export
Evidence • Schema • Steps • Receipts
Initial library
$L_0 = (S_0, R_0)$
DeepSeek Harness
Our custom orchestration
Custom tools
Search & inspect
Capability gateway
Python Controller
State machine • State isolation
DSH child sessions
Agent loop • tool calling
Cold-start initialization
Initial Skill
Initial Reference
Skill Self-Evolution
Task
Independent Evolution Tasks
original routed entries+task-specific failure history
First-attempt success: no update
failure
success after failure
A Diagnose Failure
Grounding failure
No candidate
Knowledge defect
Revise existing skill/reference
Coverage gap
Add new knowledge
B Contrast Trajectories
Successful trajectory
Failed trajectory 1
Failed trajectory 2
Revised existing skill/reference
Fail
Independent Validation
Reset environment
Fresh session
No history
Pass
Exploration
Revision
Retry unchanged
Add accepted candidate to pool P
Consolidation
After All Tasks Terminate
Frozen final library